Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI3311-WE01

Title: MEng and BEng Engineering

Geotechnics 3

Time Allowed:	2 hours	
Additional Material provided:	Formula Sheet (pg. 7 to 11)	
Materials Permitted:	None.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	Answer ALL questions. Use a separate answer book for each question. All relevant workings must be shown.

Revision:	

Q1

(a) Figure Q1a sketches an aquifer that connects an upstream water reservoir (left, Figure Q1a) with a downstream river (right, Figure Q1a) through a deep layer of gravel filled gabions of length 10m and thickness 2.2m. The water level of the reservoir is 8m whereas the water level of the river is 4m. The materials above and below the layer of gravels can be considered impermeable. Assume impermeable bedrock as your reference datum.

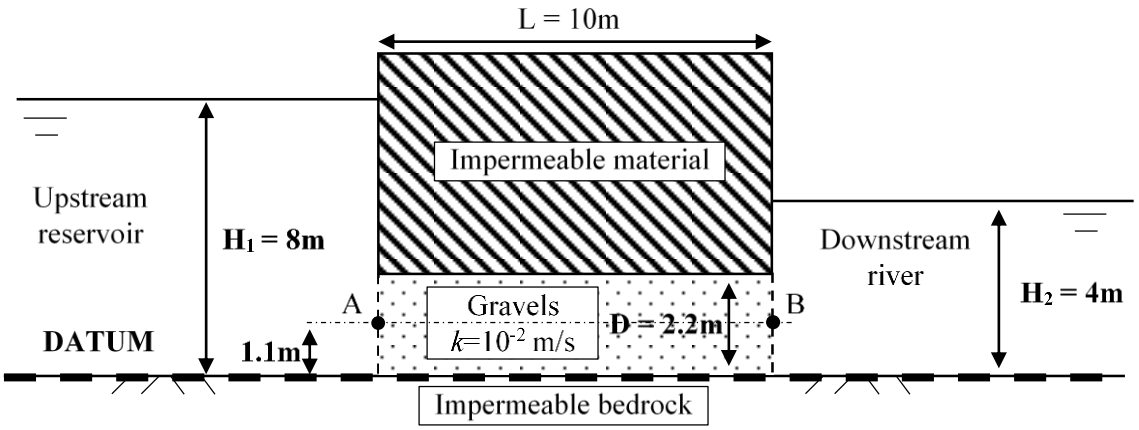


Figure Q1a

(i) Calculate the total head at points A and B. What is the direction of the flow? Justify your answer. [10%]

(ii) What is the total head of a point located at a distance of 3m from point A along the line AB in the layer of gravels? [10%]

(iii) Calculate the horizontal water flow through the layer of gravels assuming that the permeability of the gravels is $k = 10^{-2}$ m/s. [20%]

(iv) Assuming that the layer of gravels has a constant void ratio of 0.6, calculate the superficial seepage velocity and the average true velocity. Justify why their respective values are different. [20%]

continued

Q2

A cylindrical sample of a saturated clay is tested under drained triaxial conditions. At the start of the test, the pre-consolidation stress of this clay is p'_0 = 250 kPa, its overconsolidation ratio is OCR=10 and the intercept of the isotropic normal compression line is N=2.5.

(a) The sample is first subjected to an isotropic compression (from initial state A to final state B), increasing the mean effective stress from an initial value p'_A to a final p'_B = 400 kPa.

(i) Calculate the mean effective stress at A (p'_A). [10%]
(ii) Assuming that the values of specific volume at A and B are v_A=1.900 and v_B=1.757 respectively, calculate the values of lambda λ and kappa κ. [20%]
(iii) Sketch, in the v:lnp' plane, how the specific volume v varies during the isotropic loading AB (indicating the position of points A and B as well as the position of the normal compression line i.e. intercept N and gradient λ). [10%]
(iv) What is the value of the overconsolidation ratio (OCR) at point B? Is the saturated clay sample normally consolidated or overconsolidated? [10%]

(b) During the second stage of the triaxial compression test, the sample is sheared, under drained conditions, from B to the critical state at C.

(v) Calculate the value of deviatoric stress at critical state q_C assuming that M = 1.105. [20%]
(vi) Sketch, in the q:p' plane, the effective stress path during BC (indicating the position of points B and C, position and gradient of the critical state line, and the gradient of the effective stress path). [20%]
(vii) What is the value of mean effective stress at critical state p'_C? [10%]

continued

Q3

A vertical cut is excavated on a site to carry out some construction work. The site geotechnical engineer wants to ensure that the vertical cut is stable (i.e. Factor of safety of at least 1.2) when the total height excavated is H=4.5m (see Figure Q3). The material excavated is a dry silt with a unit weight γ=19kN/m^3, a friction angle φ'=30° and a cohesion c' = 15 kPa. Total major and minor principal normal stresses at failure during an unconfined undrained triaxial test are measured as σ_1=50kPa and σ_3=0kPa (=σ_2).

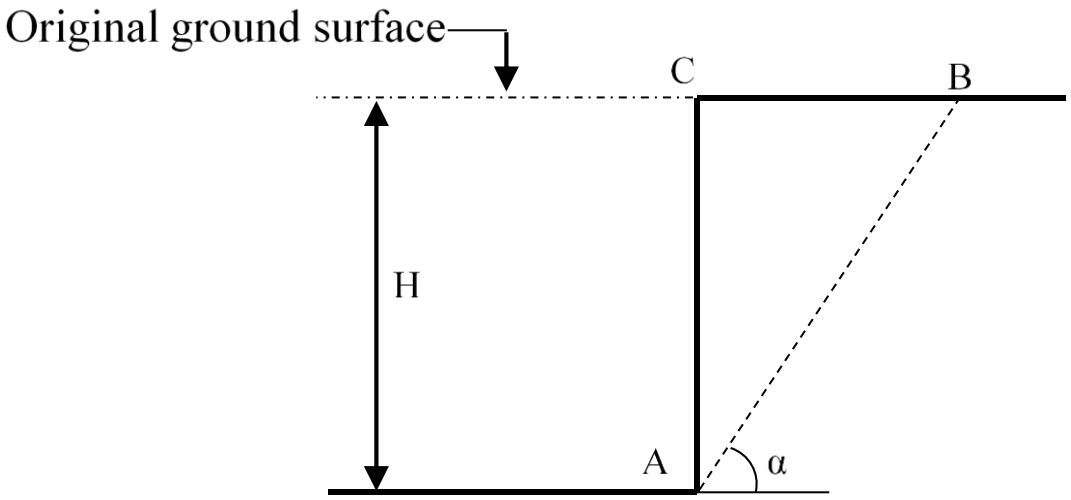


Figure Q3

Assess the stability of the cutting for α = 30°, 45°, 60° and 70°, considering the following two conditions:

(a) The excavation is only needed for temporary works. [35%]
(b) The excavation is needed for a long period of time. [25%]

(c) Based on the results obtained, discuss how the factor of safety varies with angle α. Identify in your answer which cases are more critical for high and low values of alpha, and explain the reasons for this. [40%]

continued

Q4

A square footing is founded on a saturated silty clay with characteristic angle of friction $\varphi'_{k2}=30°$, characteristic cohesion $c_{k2}'=15kPa$ and characteristic undrained shear strength $C_{uk2}=100kPa$. The layer of silty clay lies between a layer of sand and the bedrock (see Figure Q4). The footing supports a vertical permanent load $V_{1k}=2725kN$ (including self-weight) and a vertical variable load of $V_{2k}=1000kN$, both applied centrally to the footing (so that eccentricity can be ignored). The unit weight of the sand is $\gamma_{k1}=18kN/m^3$ and the unit weight of the silty clay is $\gamma_{k2}=20kN/m^3$. The water table is at the ground surface. Assume that the characteristic angle of friction of the sand is $\varphi'_{k1}=35°$.

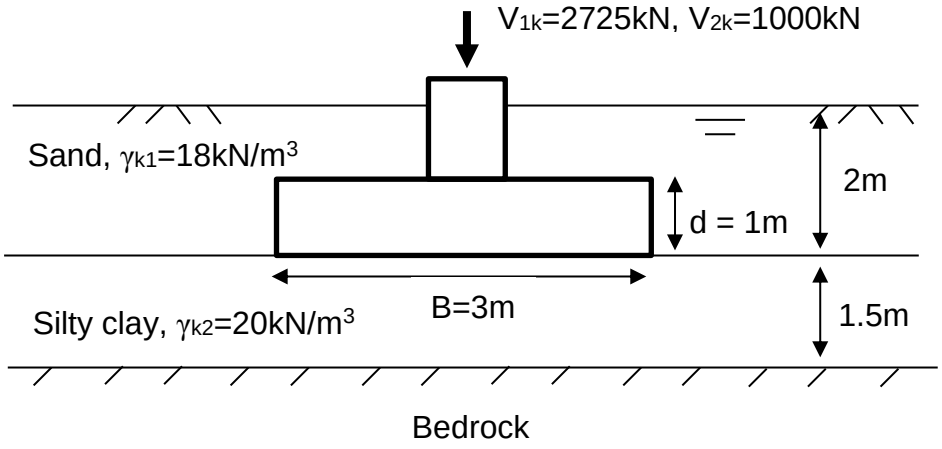


Figure Q4

Following the next steps, assess the safety of the square footing in accordance with the Eurocode 7 (Case C) for drained and undrained conditions.

(a) What is the design bearing pressure q_{Ed}? [20%]
(b) What is the design bearing resistance for drained conditions, q_{Rd} (drained)? [40%]
(c) What is the design bearing resistance for undrained conditions, q_{Rd} (undrained)? [20%]
(d) Is the footing safe? Justify your answer based on the results obtained in (b) and (c). [20%]

continued

Phase relationships

$e=\frac{V_v}{V_s}$	$S_r=\frac{V_w}{V_v}$	$v=1+e$	$n=\frac{e}{1+e}$	$w=\frac{M_w}{M_s}$
$\rho_s=G_s\rho_w$	$wG_s=S_re$	$\rho=\frac{M_s+M_w}{V_s+V_v}$	$\rho=\frac{(G_s+S_re)\rho_w}{1+e}$	$\gamma=\rho g$
$\gamma=\frac{(G_s+S_re)\gamma_w}{1+e}$	$\gamma_{sat}=\frac{(G_s+e)\gamma_w}{1+e}$	$\gamma'=\gamma_{sat}-\gamma_w$	$\rho_d=\frac{G_s\rho_w}{1+e}=\frac{\rho}{1+w}$	$\gamma_w\approx 10$ kN/m^3

Stresses and pore pressures in soils:

$\sigma_v=\rho gz=\gamma z$	$u_w=\rho_w gz_w=\gamma_w z_w$	$\sigma_v'=\sigma_v-u$	$u_a-u_w=\frac{2T}{R}$	T $\approx$ 0.07 N/m

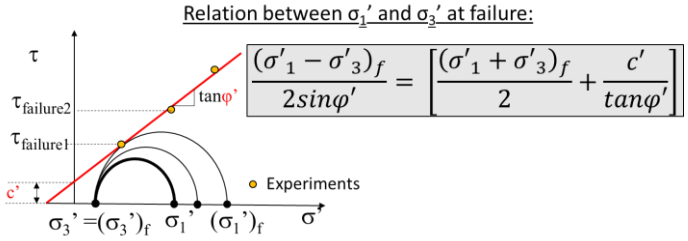
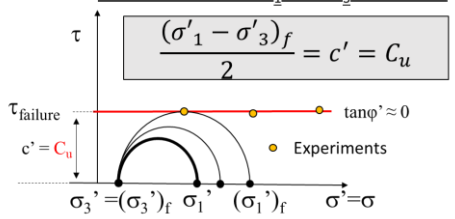
Elastic stress-strain relations:

The volumetric strain is : $\Delta\varepsilon_v=\Delta\varepsilon_1+\Delta\varepsilon_2+\Delta\varepsilon_3=\Delta\varepsilon_x+\Delta\varepsilon_y+\Delta\varepsilon_z$

$\Delta\varepsilon_x=\frac{1}{E'}\left(\Delta\sigma_x'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{xy}=\frac{\Delta\tau_{xy}}{G'}$	--
$\Delta\varepsilon_y=\frac{1}{E'}\left(\Delta\sigma_y'-\nu'\Delta\sigma_x'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{yz}=\frac{\Delta\tau_{yz}}{G'}$	$K'=\frac{E'}{3(1-2\nu')}$
$\Delta\varepsilon_z=\frac{1}{E'}\left(\Delta\sigma_z'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_x'\right)$	$\Delta\gamma_{zx}=\frac{\Delta\tau_{zx}}{G'}$	$G'=\frac{E'}{2(1+\nu')}$
$\Delta\varepsilon_v=\frac{\Delta p'}{K'}$		$\Delta\varepsilon_q^e=\frac{\Delta q}{3G'}$
$p'=\frac{\sigma_1'+\sigma_2'+\sigma_3'}{3}$		$q=\left[\frac{(\sigma_1'-\sigma_2')^2+(\sigma_2'-\sigma_3')^2+(\sigma_3'-\sigma_1')^2}{2}\right]^{1/2}$

where σ_1', σ_2' and σ_3' are the principal effective stresses (i.e. $\sigma_1'\geq\sigma_2'\geq\sigma_3'$)

Failure envelopes:

$\tau=c'+\sigma'\tan\varphi'$ (Mohr-Coulomb)	$\tau=C_u$ (Tresca)
Relation between σ_1' and σ_3' at failure: $\frac{(\sigma'_1-\sigma'_3)_f}{2\sin\varphi'}=\left[\frac{(\sigma'_1+\sigma'_3)_f}{2}+\frac{c'}{\tan\varphi'}\right]$ 	Relation between σ_1' and σ_3' at failure: $\frac{(\sigma'_1-\sigma'_3)_f}{2}=c'=C_u$ 

Ground water flow:

$h=z+\frac{u}{\rho g}=z+\frac{u}{\gamma_w}$	$q_x=Aki_x=Ak\left(-\frac{\partial h}{\partial x}\right)$	$v_x=ki_x$	$v_{true,x}=v_x\frac{1}{n}$

Constant head permeameter:	Falling head permeameter:	Pumping test:
$q=Ak\frac{h_1-h_2}{L}$	$\ln\left(\frac{h_0}{h}\right)=\frac{Ak}{aL}t$	$\left(H^2-h^2\right)=\frac{q}{\pi k}\left(\ln R_0-\ln r\right)$

Laplace's Equation (3D)	Laplace's Equation (2D, flownets)	
$\frac{\partial^2 h}{\partial x^2}+\frac{\partial^2 h}{\partial y^2}+\frac{\partial^2 h}{\partial z^2}=0$	$\frac{\partial^2 h}{\partial x^2}+\frac{\partial^2 h}{\partial y^2}+\frac{\partial^2 h}{\partial z^2}=0$	$q=k\frac{N_f}{N_d}\Delta h;\ \ u=\gamma_w(h-z)$

continued

Critical hydraulic gradient:	$i_c = \dfrac{\gamma - \gamma_w}{\gamma_w}$

1D flow through an unconfined aquifer:	1D flow through a confined aquifer:
$q_x = \dfrac{k\left(H_1^2 - H_2^2\right)}{2L}$	$q_x = Aki_x = -Ak\dfrac{\partial h}{\partial x} = -k\dfrac{H_2 - H_1}{L}D$
$h = \sqrt{\dfrac{\left(H_2^2 - H_1^2\right)}{L}x + H_1^2}$	$h = \dfrac{H_2 - H_1}{L}x + H_1$

1D straining (oedometric test conditions):

1D normal compression line:	1D swelling line:
$e = e_1 - C_c \log_{10} \sigma'_v$	$e = e_2 - C_s \log_{10} \sigma'_v$

$OCR_{1D} = \dfrac{max.\ past\ value\ of\ \sigma'_v}{current\ value\ of\ \sigma'_v} = \dfrac{\sigma'_{vc}}{\sigma'_v}$	$\sigma'_h = K_0\sigma'_v$	$K_{0nc} \approx 1 - \sin\varphi'$	$K_0 \approx \left(1 - \sin\varphi'\right)\sqrt{OCR}$

Settlement of a very wide foundation:		
$\dfrac{\Delta H}{H_0} = \dfrac{e_0 - e_1}{1 + e_0}$	$\Delta H = \dfrac{H_0}{1 + e_0}\left[C_s \log_{10}\dfrac{\sigma'_{vc}}{\sigma'_{v0}} + C_c \log_{10}\dfrac{\sigma'_{v1}}{\sigma'_{vc}}\right]$	$\Delta\varepsilon_v = m_v\Delta\sigma'_v$

1D consolidation:		
$\dfrac{\partial u_e}{\partial t} = c_v\left(\dfrac{\partial^2 u_e}{\partial z^2}\right)$	$c_v = \dfrac{k}{m_v\gamma_w}$	$T_v = \dfrac{c_v t}{d^2}$

Axisymmetric conditions (triaxial test conditions):

$\Delta\varepsilon_v = \Delta\varepsilon_a + 2\Delta\varepsilon_r$	$\Delta\varepsilon_q = \dfrac{2}{3}\left(\Delta\varepsilon_a - \Delta\varepsilon_r\right)$
$p' = \dfrac{1}{3}\left(\sigma_a' + 2\sigma_r'\right)$	$q = \sigma_a - \sigma_r$

where σ'_a and σ'_r are the principal effective axial and radial stresses respectively

Iso-normal compression line:	Iso-swelling line:
$v = N - \lambda \ln p'$	$v = v_\kappa - \kappa \ln p'$

Critical states:	
$q = Mp'$	$v = \Gamma - \lambda \ln p'$

where $\Gamma = N - (\lambda - \kappa)\ln(2)$

Relation between friction angle and critical state gradient:	$M = \dfrac{q}{p'} = \dfrac{6\sin\varphi'}{3 - \sin\varphi'}$

Overconsolidation ratio (OCR in terms of p')	$OCR = \dfrac{max.\ past\ value\ of\ p'}{current\ value\ of\ p'} = \dfrac{p'_0}{p'}$

Partial factors Eurocode 7 & factor of safety: (Bond & Harris, 2008; *Decoding EC7*)

$q_{Ed} \le q_{Rd}$ where, q_{Ed} is the design pressure and q_{Rd} is the design resistance	CASE	ACTIONS			Soil properties	
		Permanent		Variable	tan φ', c'	C_u
		Unfavourable	Favourable	Unfavourable		
	C	1.00	1.00	1.30	1.25	1.4

continued

Factor of safety, FoS:	$FoS = \frac{\textit{available shear force, } T_f}{\textit{allowable working load, } T}$; $FoS = \frac{\textit{available bearing capcity, } \tau_f}{\textit{allowable bearing capacity, } \tau}$

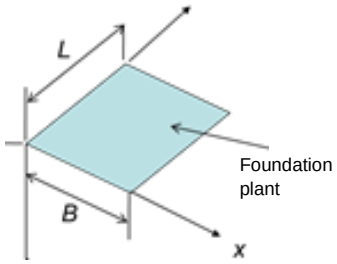
Shallow foundation settlement, δ:

Cohesionless soils:

Burland & Burbage:

$$\delta\ (mm) = f_s \cdot f_l \cdot f_t \cdot \left[\left(q'_n - \frac{2}{3}p'_0\right) \cdot B^{0.7} \cdot I_c\right]$$

where,



$f_s = \left(\frac{1.25 \cdot L/B}{\frac{L}{B} + 0.25}\right)^2$	$f_l = \frac{H}{z_l}\left(2 - \frac{H}{z_l}\right)$	$f_t = 1 + R_3 + R \cdot log\left(\frac{t}{3}\right)$

Cohesive soils: $\delta = \delta_i + \delta_c$ where,

$\delta_i\ (mm) = \mu_1 \cdot \mu_0 \cdot q_n \frac{B}{E_u}$	$\delta_c\ (mm) = \mu_g \cdot \delta_{oed}$ where, $\delta_{oed}\ (mm) = m_v \cdot \sigma'_v \cdot H$

Bearing capacity:

Brinch-Hansen (drained all soils)

$$q_f = c' \cdot N_c \cdot s_c + p'_{ov} \cdot N_q \cdot s_q + \frac{1}{2} \cdot \gamma' \cdot B \cdot N_y \cdot s_y$$

where,

$N_q = e^{\pi \tan \varphi'} \cdot \left(\tan(45 + \varphi'/2)\right)^2$	$s_q = 1 + (B'/L') \cdot \sin \varphi'$
$N_c = \left(N_q - 1\right) \cdot \cot \varphi'$	$s_c = \left(s_q \cdot N_q - 1\right)/\left(N_q - 1\right)$
$N_y = 2 \cdot \left(N_q - 1\right) \cdot \tan \varphi'$	$s_y = 1 - 0.3 \cdot (B'/L')$

depth factors are not permitted within EC7; load inclination, base and ground factors ignored.

Fine-grained soils (undrained):

$q_f = 5.14 \cdot C_u \cdot s'_c$ where,

$$s'_c = 1 + 0.2 \cdot \left(\frac{B'}{L'}\right)$$

load inclination, base and ground factors are ignored.

Bearing capacity in piles:

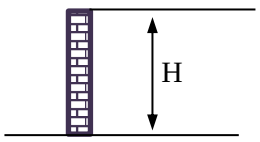
$$Q_u = Q_s + Q_b - W_p$$

Cohesionless soils:	Cohesive soils:
$Q_s = q_s \cdot A_s = A_s \cdot k_s \cdot \sigma_{v0} \cdot \tan \delta$	$Q_s = q_s \cdot A_s = \alpha \cdot C_u \cdot A_s$
$Q_b = q_b \cdot A_b = N_q \cdot \sigma'_{v0} \cdot A_b$	$Q_b = q_b \cdot A_b = N_c \cdot C_{ub} \cdot A_b$

Passive and active pressures:

$\sigma'_h = K_a \cdot \sigma'_v$ (active)	$\sigma'_h = K_b \cdot \sigma'_v$ (passive)

RANKINE'S theory: H high of the layer of soil behind the wall retaining wall, φ' is the friction angle and c' is the cohesion



In the following, $P_a = P_a' + U$ and $P_p = P_p' + U$,

where U is the resultant of the pore water pressure.

$K_a = \tan^2\left(\frac{\pi}{4} - \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1+\sin\varphi'}\right)^2 = \frac{1-\sin\varphi'}{1+\sin\varphi'}$	$K_p = \tan^2\left(\frac{\pi}{4} + \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1-\sin\varphi'}\right)^2 = \frac{1+\sin\varphi'}{1-\sin\varphi'}$

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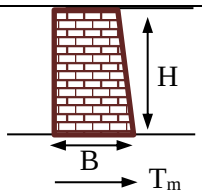
Cohesionless soils:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u$	$\sigma_h' = \sigma_v' K_a - 2c'\sqrt{K_a}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{\gamma}{2}\left(H - \frac{2C_u}{\gamma} \right)^2$	$P_a' = \frac{K_a \gamma'}{2}\left(H - \frac{2c'}{\gamma'\sqrt{K_a}} \right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = 2C_u + \sigma_v$	$\sigma_h' = \sigma_v' K_p + 2c'\sqrt{K_p}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}\left(\gamma H + 4C_u \right)$	$P_p' = \frac{1}{2} K_p \gamma' H^2 + 2c'\sqrt{K_p} H$

<u>COULOMB'S theory:</u> Friction and adhesion between soil and wall are φ_w and c_w, resp.

$$K_a = \left(\frac{\cos\varphi'}{1 + \sin\varphi' \sqrt{1 + \tan\varphi_w' / \tan\varphi'}} \right)^2 \frac{1}{\cos\varphi_w'} \qquad K_p = \left(\frac{\cos\varphi'}{1 - \sin\varphi' \sqrt{1 + \tan\varphi_w' / \tan\varphi'}} \right)^2 \frac{1}{\cos\varphi_w'}$$

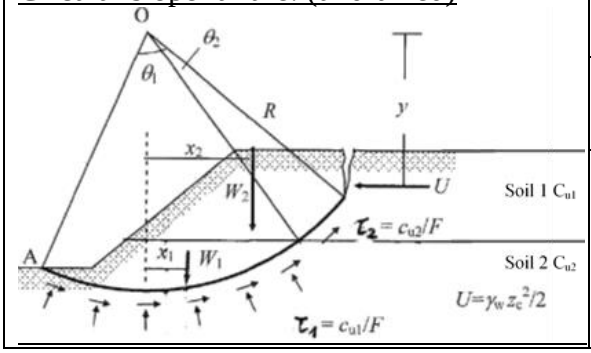
Cohesionless:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u\sqrt{1 + \frac{C_w}{C_u}}$	$\sigma_h' = K_{a1}\sigma_v' - c' K_{a2}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{\gamma}{2}\left(H - \frac{2C_u}{\gamma}\sqrt{1 + \frac{C_w}{C_u}} \right)^2$	$P_a' = \frac{K_{a1}\gamma'}{2}\left(H - \frac{c' K_{a2}}{\gamma' K_{a1}} \right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = \gamma z + 2C_u\sqrt{1 + \frac{C_w}{C_u}}$	$\sigma_h' = \gamma z K_{p1} + c' K_{p2}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}\left(\gamma H + 4C_u\sqrt{1 + \frac{C_w}{C_u}} \right)$	$P_p' = \frac{1}{2} K_{p1} \gamma' H^2 + c' K_{p2} H$

where,

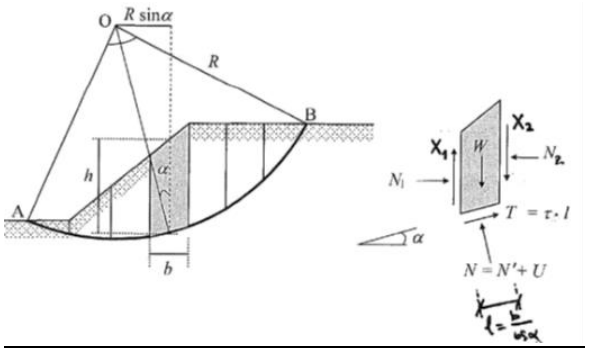
$K_{a1} = K_a$	$K_{p1} = K_p$	
$K_{a2} = \frac{\cos\varphi'\left(2\sqrt{1 + c_w'/c'} + \left(2 + c_w'/c'\right)\sin\varphi' \right)}{\left(1 + \sin\varphi' \sqrt{1 + \tan\varphi_w'/\tan\varphi'} \right)^2 \cos\varphi_w'}$	$K_{p2} = \frac{\cos\varphi'\left(2\sqrt{1 + c_w'/c'} - \left(2 + c_w'/c'\right)\sin\varphi' \right)}{\left(1 - \sin\varphi' \sqrt{1 + \tan\varphi_w'/\tan\varphi'} \right)^2 \cos\varphi_w'}$	
<u>Cantilever wall in sand:</u>	$FoS = \frac{N' \tan\varphi_w'}{T_m} = \frac{(W - U)\tan\varphi_w'}{T_m}$	
<u>Gravity wall in sand:</u>	$FoS = \frac{N \tan\varphi_w'}{T_m}$	
<u>Gravity wall in clay:</u> 	(Undrained) $FoS = \frac{C_w B}{T_m}$	(Drained) $FoS = \frac{N' \tan\varphi_w' + c_w' B}{T_m}$

continued

Slope stability:

Infinite slope with parallel flow: H: layer thickness, m·H: position of water table from bedrock, β: slope angle; γ_1 (dry soil), γ_2 (saturated soil).	$FoS = \dfrac{c' + H\left(\gamma_1(1-m) + m(\gamma_2 - \gamma_w)\right)\cos^2\beta\tan\varphi'}{H\left((1-m)\gamma_1 + m\gamma_2\right)\cos\beta\sin\beta}$
Vertical cut (dry soil):	$FoS = \dfrac{4C_u}{H\gamma\sin 2\alpha}$ (undrained)
	$FoS = \dfrac{4(c' + \sigma'\tan\varphi')}{H\gamma\sin 2\alpha}$ (drained)
Circular slope failure: (undrained) 	$FoS = \dfrac{R^2\theta C_u}{Wx}$
	$FoS = \dfrac{R^2\theta C_u}{Wx + Uy}$ (with cracks)
	$FoS = \dfrac{(\theta_1 C_{u1} + \theta_2 C_{u2})R^2}{W_1 x_1 + W_2 x_2 + Uy}$ (layered soil)

$\sin 2\alpha = 2\sin\alpha\cos\alpha$, $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$ and arc-length: $AB = R\theta$ (θ is in radians).

Circular: (drained): 	$FoS = \dfrac{\sum_{i=1}^{n} l\left(c' + \sigma'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} = \dfrac{\sum_{i=1}^{n}\left(lc' + N'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$
Modified Bishop's b: width of the slice, l: length of the slice, α: inclination of the slice to the horizontal direction.	$FoS = \dfrac{\sum_{i=1}^{n}\left(bc' + \left(W - U\cos\alpha\right)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} M(\alpha)$ where, $M(\alpha) = \dfrac{1}{\cos\alpha\left[1 + \dfrac{\tan\alpha\tan\varphi'}{FoS}\right]}$
Fellenius:	$FoS = \dfrac{\sum_{i=1}^{n}\left(lc' + \left(W\cos\alpha - ul\right)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$

END